

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
FACULTY OF TECHNOLOGY & ENGINEERING
Chandubhai S. Patel Institute of Technology - Changa
Department of Information Technology

Subject Name: Operating System

Subject Code: IT343

Semester: V

Academic year: 2019-20

Practical List

Instructions:

ISO Practical Format: Aim, Software/Hardware Required, Knowledge Required, Theory, Algorithm/Flow chart, Program, Input and Output, Advantages and Disadvantages, Conclusion, Questions and Answers.

Sr.No.	Aim of the Practical																						
1.	Study Practical: - UNIX Architecture - Types of OS - Flavors of LINUX																						
2.	Study following Unix commands with option: <table border="1"> <tr> <td>User Access:</td><td>login, logout, passwd, exit</td></tr> <tr> <td>Help:</td><td>man, help</td></tr> <tr> <td>Directory:</td><td>mkdir, rmdir, cd, pwd, ls</td></tr> <tr> <td>Editor:</td><td>vi, gedit, ed, sed</td></tr> <tr> <td>File Handling / Text Processing:</td><td>cp, mv, rm, sort, cat, pg, lp, pr, file, find, more, cmp, diff, comm, head, tail, cut, grep, touch, tr, uniq</td></tr> <tr> <td>Security and Protection:</td><td>chmod, chown, chgrp, newgrp</td></tr> <tr> <td>Information:</td><td>learn, man, who, date, cal, tty, calendar, time, bc, whoami, which, hostname, history, wc</td></tr> <tr> <td>System Administrator:</td><td>su or root, adduser, rmuser</td></tr> <tr> <td>Terminal:</td><td>echo, printf, clear</td></tr> <tr> <td>Process:</td><td>ps, kill, exec</td></tr> <tr> <td colspan="2">I/O Redirection (<, >, >>), Pipe(), *, gcc</td></tr> </table>	User Access:	login, logout, passwd, exit	Help:	man, help	Directory:	mkdir, rmdir, cd, pwd, ls	Editor:	vi, gedit, ed, sed	File Handling / Text Processing:	cp, mv, rm, sort, cat, pg, lp, pr, file, find, more, cmp, diff, comm, head, tail, cut, grep, touch, tr, uniq	Security and Protection:	chmod, chown, chgrp, newgrp	Information:	learn, man, who, date, cal, tty, calendar, time, bc, whoami, which, hostname, history, wc	System Administrator:	su or root, adduser, rmuser	Terminal:	echo, printf, clear	Process:	ps, kill, exec	I/O Redirection (<, >, >>), Pipe(), *, gcc	
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3.	- Write a script called hello which outputs the following: - your username - the time and date - who is logged on? - Also output a line of asterices (*****) after each section. - Write a shell program to find the largest integer among the three integers given as arguments. - Write a shell script to simulate a simple calculator.																						

	4. Write a shell script to sort the number in ascending order and also calculate the shell script run time.(Using array).
4.	- Write a shell program to count the following in a text file. - Number of vowels in a given text file. - Number of blank spaces. - Number of characters. - Number of symbols. - Number of lines - Write a shell script which will take a file name from the user and finds that whether the file is there or not in a current working directory and displays the appropriate message. - Write a shell script which compares two files given by the user and if both files are same then delete the second one, if not then merge the two files in a new file.
5.	Write a C program to list for every file in a directory, its inode number and file name.
6.	Write a C program in UNIX to implement Process scheduling algorithms and compare. - First Come First Serve (FCFS) Scheduling - Shortest-Job-First (SJF) Scheduling
7.	Write a C program in UNIX to implement inter process communication (IPC) using Semaphore.
8.	Write a C program in UNIX to implement Bankers algorithm for Deadlock Avoidance.
9.	Write a C program in UNIX to solve dining philosophers' problem.
10.	Write a C program in UNIX to perform Memory allocation algorithms and calculate Internal and External Fragmentation. (First Fit, Best Fit, Worst Fit).